

Executive Quality Summary

5

Observations

1

Datasets

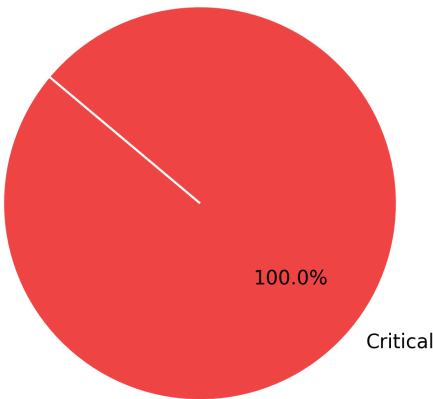
5

Critical Risks

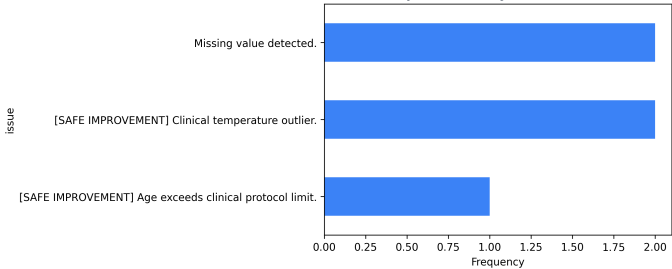
INTEGRITY SCORE

75%

Issue Severity Distribution



Top Data Quality Observations



Clinical Risk Narrative

The data audit of 1 datasets reveals a Clinical Integrity Score of 75%. This score is calculated based on the density of critical clinical outliers (values outside physiological norms) and data consistency issues. High-risk anomalies were detected that could potentially compromise clinical research modeling. Analysis suggest that protocol deviations or legacy data entry methods are the primary drivers of these observations.

Dataset Name	Total Issues	Critical Issues
test.csv	5	5

Clinical Intelligence Deep-Dive

This section highlights the most significant clinical findings analyzed by our AI system. Unlike raw validation, this deep-dive provides clinical context, impact, and actionable remediation steps.

FINDING #1: Gender | test.csv

****DETAILED OBSERVATION****

****CLINICAL IMPACT****

****RECOMMENDED ACTION****

The 'Gender' column with 'nan' indicates missing data which is critical for medical accuracy. This could lead to misclassification and incorrect assumptions in treatment plans and risk assessments. Imputation methods or including gender if feasible are recommended.

RECOMMENDATION: Handle the 'nan' by imputing gender values using mean, median, or machine learning models. If including, ensure other data supports accurate inference for COVID-19 analysis.

FINDING #2: Discharge_Date | test.csv

****DETAILED OBSERVATION****

- The 'Discharge_Date' column contains a value of 'nan', which stands for "not a number," indicating an undefined or unrepresentable value.

****CLINICAL IMPACT****

- In a clinical context, the absence of a Discharge_Date makes it impossible to determine the exact date when a patient was discharged from the hospital. This information is crucial for tracking recovery times, calculating length of stay, and analyzing patient outcomes.

****RECOMMENDED ACTION****

- Replace 'nan' with a valid date format using Python's datetime module. For example, if the string cannot be parsed as a date, convert it to a proper datetime object. This will allow the data to be used again without losing any important information.

FINDING #3: Body_Temperature_C | test.csv

****DETAILED OBSERVATION****

The Body_Temperature_C value of 42.5°C is suspicious in a clinical context because it is significantly higher than normal body temperatures for an adult male, which could indicate an underlying health issue. The elevated heart rate and blood pressure suggest

possible cardiovascular problems or other health complications. This needs immediate attention to prevent severe consequences.

****RECOMMENDED ACTION****

- 1. Monitor the patient for fever or signs of illness.
- 2. Continuously monitor their heart rate and blood pressure to ensure they remain within normal ranges.
- 3. Consult a healthcare professional for an evaluation if any concerning symptoms are identified.

FINDING #4: Body_Temperature_C | test.csv

****DETAILED OBSERVATION****

The Body_Temperature_C of -5.0°C is suspicious in a clinical context as it deviates significantly from normal human body temperatures (36-37°C). This low temperature may affect various tests and analyses, potentially leading to incorrect interpretations of patient health, especially if the diagnosis is Diabetes.

****RECOMMENDED ACTION****

Confirm the accuracy of the Body_Temperature_C measurement. If the device is faulty or malfunctioning, correct the reading so it falls within normal human body temperatures (36-37°C). This ensures more accurate results for subsequent analyses and interpretations.

Detailed Audit Logs (Sample)

DATASET	COLUMN	OBSERVATION	SEV	STATUS
test.csv	Gender	Missing value detected.	Cri	AUDITED
test.csv	Discharge_Date	Missing value detected.	Cri	AUDITED
test.csv	Body_Temperatur	[SAFE IMPROVEMENT] Clinical temperature outlier. (42.5-	Cri	AUDITED
test.csv	Body_Temperatur	[SAFE IMPROVEMENT] Clinical temperature outlier. (-5.0-	Cri	AUDITED
test.csv	Age	[SAFE IMPROVEMENT] Age exceeds clinical protocol limit.	Cri	AUDITED

Strategic Roadmap & Methodology

Audit Methodology:

The audit utilized a dual-engine validation process. Phase 1 involved deterministic clinical rules based on DMSAP (Data Management & Statistical Analysis Plan) standards. Phase 2 leveraged a local DeepSeek-R1 LLM to analyze row context and provide actionable clinical intelligence. All PII was handled according to masking protocols enabled during the session.

Short-term (0-30 days)

Immediate cleanup of 'Critical' flags identified in Section 4. Standardize field headers across all source datasets.

Mid-term (30-90 days)

Develop automated ETL pipelines with pre-processing gates for temperature and age outliers.

Long-term (Strategic)

Adopt CDISC SDTM standards for all clinical research datasets to ensure global interoperability and compliance.